

Understanding difference-in-difference methods by simulation and example

Dr. David M. Drukker

The Wilbur O. and Ann Powers College of Business
Clemson University

31 October 2024

Overview

- 1 Review of cross-sectional causal inference
- 2 Two-period DID
- 3 RA, IPW, and AIPW estimators
- 4 Linked appendices

Review of cross-sectional causal inference

The difference-in-difference (DID) estimator is a complicated simple estimator

- In many applications, DID estimators are implemented using a simple regression strategy
- But understanding,
 - what a DID estimator is really trying to estimate
 - when a simple regression strategy worksrequires concepts from causal inference and some complicated conditional-expectation calculations
- Furthermore, it's not easy to see how the conditions required for DID relate to underlying models

Models and simulations for understanding

- The conditions required by modern causal-inference methods are non-parametric
 - They don't use specific parametric forms for the underlying processes
- Explicit parametric models can help us understand these modern conditions
- Simulation techniques that draw from explicit parametric models offer a hands-on way to gain some understanding of the concepts used by DID and to see what works and what does not work
- This talk uses explicit models and simulations in Stata to illustrate the concepts discussed

Need some background from causal inference

- DID estimators are designed to estimate something called an average treatment effect on the treated (ATET) from panel data or repeated cross-sections of data
- Let's review some things from causal inference for cross-sectional data to see what an ATET is and what are the challenges and opportunities that panel data or repeated cross-sectional data bring

Potential outcomes

- Potential outcomes are the data that we wish we had
 - But we will never have this data, it's not feasible
- Consider the effect of a student-retention program in a university
 - In the spring semester, some students get the program, some students don't get the program
 - In the following fall semester, each student either enrolls or does not enroll
 - For each student,
 - there is a potential enrollment outcome when the student **gets** the program
 - there is a potential enrollment outcome when the student **does not get** the program

- We wish that we could observe both potential outcomes for each student, as in the fictional data below
- **enroll_program** indicates whether a student enrolled after getting the program
- **enroll_noprogram** indicates whether a student enrolled after not getting the program

```
. use progouts  
. list id enroll_*, abbreviate(20)
```

	id	enroll_program	enroll_noprogram
1.	1	1	0
2.	2	0	0
3.	3	1	1
4.	4	0	1
5.	5	1	1
6.	6	0	0
7.	7	1	0
8.	8	0	0
9.	9	1	1
10.	10	0	1

- In real data, we never see both potential outcomes, a problem called “the fundamental problem of causal inference” by [Holland \(1986\)](#)

- In real data, we would see the `id`, an indicator for whether the student enrolled in the program, and the observed potential outcome corresponding to value of the program indicator

```
. use progouts  
. list id program enroll , abbreviate(20)
```

	id	program	enroll
1.	1	0	0
2.	2	1	0
3.	3	0	1
4.	4	1	0
5.	5	0	1
6.	6	1	0
7.	7	0	0
8.	8	1	0
9.	9	0	1
10.	10	1	0

- Let d_i be the indicator for treatment

$$d_i = \begin{cases} 1 & \text{if get treatment} \\ 0 & \text{if not get treatment} \end{cases}$$

- Let the treated potential outcome be $y_{1,i}$
- Let the not treated (control) potential outcome be $y_{0,i}$
- Let the observed outcome be y_i
- Then

$$y_i = d_i y_{1,i} + (1 - d_i) y_{0,i}$$

```
. use progouts
. list id program enroll_* enroll , abbreviate(20)
```

	id	program	enroll_program	enroll_noprogram	enroll
1.	1	0	1	0	0
2.	2	1	0	0	0
3.	3	0	1	1	1
4.	4	1	0	1	0
5.	5	0	1	1	1
6.	6	1	0	0	0
7.	7	0	1	0	0
8.	8	1	0	0	0
9.	9	0	1	1	1
10.	10	1	0	1	0

- If we could observe both $y_{1,i}$ and $y_{0,i}$ instead of just y_i
 - There would be no need for this talk, or causal inference methods
 - Causal inference methods are about how we can use y_i to reach conclusions about functions of $y_{1,i}$ and $y_{0,i}$

Parameters of interest

$$ATE = \mathbf{E}[y_{i,1} - y_{i,0}]$$

$$ATET = \mathbf{E}[y_{i,1} - y_{i,0} | d_i = 1]$$

- The ATE is the mean of the differences between the treated potential outcome and control potential outcome
 - Measures program effect in the whole population
- The ATET is the mean of the differences between the treated potential outcome and control potential outcome, among those who get the treatment
 - Measures program effect in the subpopulation that gets the program

Assumptions

- Because we only see the observed outcome, and not the potential outcomes, we need some assumptions to estimate the ATE or the ATET
- The assumptions that we need to estimate ATET are weaker (more likely to hold) than the assumptions that we need to estimate the ATE

Assumptions needed

- In the cross-sectional framework, we need three types of assumptions
 - CMI** The conditional mean-independence CMI assumption restricts the dependence between the treatment-assignment process and the potential outcomes
 - Overlap** The overlap assumption ensures that each individual could get given treatment levels with positive probability
 - IID** The independent-and-identically-distributed (IID) sampling assumption ensures that the potential outcomes and treatment status of each individual are unrelated to the potential outcomes and treatment statuses of all the other individuals in the population
- In practice, we assume independent but not identically distributed observations, not IID
- Frequently, have independence over panels or clusters

Identification of ATE

- In causal inference, an identification argument shows that we can recover the desired function of the potential outcomes from the observable data in the population
- Given identification, estimators then use sample versions of the population functions to estimate the desired function of the potential outcomes
- Identification arguments are usually non parametric, they don't impose a particular functional form for the conditional expectations (regression functions) and conditional probabilities (probits/logits) involved
- In practice, estimators usually use explicit functional forms for the conditional expectations and conditional probabilities required

CMI assumption

- CMI assumptions play a key role causal inference
 - In econometric terms, CMI assumptions are the exogeneity assumptions
- Requiring that

$$\begin{aligned}\mathbf{E}[y_{1,i}|\mathbf{x}_i, d_i] &= \mathbf{E}[y_{1,i}|\mathbf{x}_i] \\ \mathbf{E}[y_{1,i}|\mathbf{x}_i, d_i = 0] &= \mathbf{E}[y_{1,i}|\mathbf{x}_i, d_i = 1]\end{aligned}$$

means that there are no variables besides $\mathbf{x}$ that affect both d_i and $y_{1,i}$

- In a modeling context, suppose that

$$\begin{aligned}y_{1,i} &= \mathbf{x}_i\boldsymbol{\beta} + \epsilon_i \\ d_{1,i} &= \mathbf{x}_i\boldsymbol{\gamma} + \eta_i > 0\end{aligned}$$

Then CMI means that η_i and ϵ_i are independent

Identification of ATE from observed outcome

- Key assumptions:

- 1 CMI of $y_{0,i}$: $\mathbf{E}[y_{0,i}|\mathbf{x}_i, d_i] = \mathbf{E}[y_{0,i}|\mathbf{x}_i]$
 $\mathbf{E}[y_{0,i}|\mathbf{x}_i, d_i = 0] = \mathbf{E}[y_{0,i}|\mathbf{x}_i, d_i = 1] = \mathbf{E}[y_{0,i}|\mathbf{x}_i]$
- 2 CMI of $y_{1,i}$: $\mathbf{E}[y_{1,i}|\mathbf{x}_i, d_i] = \mathbf{E}[y_{1,i}|\mathbf{x}_i]$
 $\mathbf{E}[y_{1,i}|\mathbf{x}_i, d_i = 0] = \mathbf{E}[y_{1,i}|\mathbf{x}_i, d_i = 1] = \mathbf{E}[y_{1,i}|\mathbf{x}_i]$
- 3 Overlap ATE: $\epsilon < \Pr(d_i = 1|\mathbf{x}_i) < 1 - \epsilon$

Recall that $y_i = d_i y_{1,i} + (1 - d_i) y_{0,i}$

$$\begin{aligned}\mathbf{E}[y_i|\mathbf{x}_i, d_i] &= d_i \mathbf{E}[y_{1,i}|\mathbf{x}_i, d_i] + (1 - d_i) \mathbf{E}[y_{0,i}|\mathbf{x}_i, d_i] \\ &= d_i \mathbf{E}[y_{1,i}|\mathbf{x}_i] + (1 - d_i) \mathbf{E}[y_{0,i}|\mathbf{x}_i] \quad (\text{by CMI})\end{aligned}$$

$$\begin{aligned}\text{which yields} \quad \mathbf{E}[y_i|\mathbf{x}_i, d_i = 1] &= \mathbf{E}[y_{1,i}|\mathbf{x}_i] \\ \mathbf{E}[y_i|\mathbf{x}_i, d_i = 0] &= \mathbf{E}[y_{0,i}|\mathbf{x}_i]\end{aligned}$$

- Iterated expectation produces

$$\begin{aligned}\mathbf{E}[y_{1,i}] &= \mathbf{E}[\mathbf{E}[y_{1,i}|\mathbf{x}_i]] \\ \mathbf{E}[y_{0,i}] &= \mathbf{E}[\mathbf{E}[y_{0,i}|\mathbf{x}_i]]\end{aligned}$$

RA estimator

The identification of the ATE from the observed outcome suggests the RA estimator

Because $\mathbf{E}[y_{1,i} - y_{0,i}] = \mathbf{E}[y_{1,i}] - \mathbf{E}[y_{0,i}]$

$$\mathbf{E}[y_{1,i}] = \mathbf{E}[\mathbf{E}[y_{1,i}|\mathbf{x}_i]] = \mathbf{E}[\mathbf{E}[y|\mathbf{x}_i, d_i = 1]]$$

$$\mathbf{E}[y_{0,i}] = \mathbf{E}[\mathbf{E}[y_{0,i}|\mathbf{x}_i]] = \mathbf{E}[\mathbf{E}[y|\mathbf{x}_i, d_i = 0]]$$

The RA estimator for the ATE does the following

- 1 It uses the treated observations and a regression model to estimate $\mathbf{E}[y|\mathbf{x}_i, d_i = 1]$
 - Sample mean of predictions ($\widehat{y1}_m$) estimates $\mathbf{E}[y_{1,i}]$
- 2 It uses the control observations and a regression model to estimate $\mathbf{E}[y|\mathbf{x}_i, d_i = 0]$
 - Sample mean of predictions ($\widehat{y0}_m$) estimates $\mathbf{E}[y_{0,i}]$
- 3 Use $\widehat{y1}_m - \widehat{y0}_m$ to estimate the ATE

Nonparametric identification versus parametric estimation

- Note the identification argument for the ATE does not assume a specific functional form for the regression functions $E[y|\mathbf{x}_i, d_i = 0]$, $E[y|\mathbf{x}_i, d_i = 1]$
 - For this reason the identification is nonparametric
- In practice, estimating the regression nonparametrically requires large amounts of data for even a few covariates, and we tend to have many covariates
- In practice, the RA estimator uses simple, flexible functional forms in the regression models for $E[y|\mathbf{x}_i, d_i = 0]$, $E[y|\mathbf{x}_i, d_i = 1]$ see for example [Cattaneo et al. \(2013\)](#) and Methods and formulas in `teffects aipw` in [Stata Causal Inference and Treatment-Effects Estimation](#)

Conditional mean independence

- The CMI assumptions
 - $\mathbf{E}[y_{0,i} | \mathbf{x}_i, d_i = 0] = \mathbf{E}[y_{0,i} | \mathbf{x}_i]$
 - $\mathbf{E}[y_{1,i} | \mathbf{x}_i, d_i = 1] = \mathbf{E}[y_{1,i} | \mathbf{x}_i]$are general, but not necessarily intuitive
- Let's write down a simple model that illustrates when the CMI assumptions hold and when they do not hold
- Consider

$$y_{0,i} = \mathbf{x}_i \beta_0 + \epsilon_{0,i}$$

$$y_{1,i} = \mathbf{x}_i \beta_1 + \epsilon_{1,i}$$

$$d_{1,i} = \begin{cases} 1 & \text{if } \mathbf{x}_i \gamma + \eta_i \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- The two CMI assumptions hold if $\epsilon_{0,i}$ is independent of η_i and if $\epsilon_{1,i}$ is independent of η_i

- Let's generate some data from a data-generating process (DGP) that satisfies the CMI conditions
- Then do RA using the `teffects` command and by hand

An aside on random numbers

- Stata has many functions for generating random numbers
- Some examples
 - `rnormal()` generates standard normal random numbers
 - `rlogistic()` generates standard logistic random numbers
 - `rchi2(q)` generates $\chi^2(q)$ random numbers
 - Recall mean of $\chi^2(q)$ is q and variance is $2(q)$
- Remember to use `set seed #` to set the seed of the random number generator to make your results reproducible
- Let ξ_q be a $\chi^2(q)$ random process
 - $\varepsilon_q = (\xi_q - q)/\sqrt{2q}$ is a mean zero, variance 1 random variable
 - As q gets large, the distribution of ε_q gets closer to a standard normal distribution
 - For smaller q , ε_q is skewed with a long right tail

Listing 1: atesim1.do

```
1  clear all
2  // The larger the sample size, the closer to convergence
3  local n = 10000
4  local df = 25
5
6  local dist " (rchi2(`df') - `df')/sqrt(2*`df') "
7
8  set obs `n'
9  generate q = `dist'
10 generate x1 = (q + `dist')/sqrt(2) + .2
11 generate x2 = (q + `dist')/sqrt(2) + .2
12
13 generate y_0 = 1 + .5*x1 + .6*x2 + `dist'
14 generate y_1 = 1.3 + .7*x1 + .8*x2 + `dist'
15 generate d = -0.7 + .6*x1 + .8*x2 + rlogistic() > 0
16 generate y = d*y_1 + (1-d)*y_0
```

Generate data and compute ATE in sample from potential outcomes

- True ATE = $.3 + .2(.2) + .2(.2) = .38$

```
. drop _all
. set seed 12345
. quietly do atesim1
. describe
```

Contains data

```
Observations:      10,000
Variables:           7
```

Variable name	Storage type	Display format	Value label	Variable label
q	float	%9.0g		
x1	float	%9.0g		
x2	float	%9.0g		
y_0	float	%9.0g		
y_1	float	%9.0g		
d	float	%9.0g		
y	float	%9.0g		

Sorted by:

Note: Dataset has changed since last saved.

```
. generate double te = y_1-y_0
. mean te
```

Mean estimation

Number of obs = 10,000

	Mean	Std. err.	[95% conf. interval]	
te	.3705722	.014565	.342022	.3991225

Use teffects to estimate ATE by RA

```
. teffects ra (y x1 x2) (d)
Iteration 0: EE criterion = 5.915e-31
Iteration 1: EE criterion = 2.654e-32
Treatment-effects estimation      Number of obs      =      10,000
Estimator      : regression adjustment
Outcome model  : linear
Treatment model: none
```

y	Coefficient	Robust std. err.	z	P> z	[95% conf. interval]	
ATE d (1 vs 0)	.390398	.0233737	16.70	0.000	.3445863	.4362096
POmean d 0	1.237848	.0174	71.14	0.000	1.203745	1.271951

RA by hand

- Consider model ATET1 that violates the CMI conditions for the ATE

$$\begin{aligned}
 y_{0,i} &= \mathbf{x}_i \boldsymbol{\beta}_0 + \epsilon_{0,i} \\
 y_{1,i} &= \mathbf{x}_i \boldsymbol{\beta}_1 + (\epsilon_{1,i} + \eta_i) / \sqrt{1 + \sigma_\eta} \\
 d_{1,i} &= \begin{cases} 1 & \text{if } \mathbf{x}_i \boldsymbol{\gamma} + \eta_i \geq 0 \\ 0 & \text{otherwise} \end{cases}
 \end{aligned} \tag{1}$$

- Unobservables that affect treatment assignment affect treated-outcome payoff but not control-outcome payoff
- Imagine an jobs program in which participants have unobserved information about what will happen if they attend, but this information has no effect on the do-not-attend outcome
- Correlating the unobserved errors for d and $y_{1,i}$ violates the CMI that $\mathbf{E}[y_{1,i} | \mathbf{x}_i, d_i] = \mathbf{E}[y_{1,i} | \mathbf{x}_i]$
- Fortunately, identifying the ATET only requires
 - 1 CMI of $y_{0,i}$: $\mathbf{E}[y_{0,i} | \mathbf{x}_i, d_i] = \mathbf{E}[y_{0,i} | \mathbf{x}_i]$
 - 2 Overlap ATET: $Pr(d_i = 1 | \mathbf{x}_i) < 1 - \epsilon$
 Note, DO NOT require CMI of $y_{1,i}$: $\mathbf{E}[y_{1,i} | \mathbf{x}_i, d_i] = \mathbf{E}[y_{1,i} | \mathbf{x}_i]$

- For the model ATET1

$$y_{0,i} = \mathbf{x}_i \boldsymbol{\beta}_0 + \epsilon_{0,i}$$

$$y_{1,i} = \mathbf{x}_i \boldsymbol{\beta}_1 + (\epsilon_{1,i} + \eta_i) / \sqrt{1 + \sigma_\eta}$$

$$d_{1,i} = \begin{cases} 1 & \text{if } \mathbf{x}_i \boldsymbol{\gamma} + \eta_i \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

we can estimate the ATET, but not the ATE, without correcting for the endogeneity

ATET RA estimator

The RA estimator for the ATET does the following

- ① It uses the control observations and a regression model to estimate $\mathbf{E}[y_i | \mathbf{x}_i, d_i = 0]$
 - Let $\widehat{y0}_{m1}$ be the sample mean of the predicted values from this regression in the subsample for which $d_i = 1$
 - Note that this is an out-of-sample prediction
- ② It computes the sample mean for y_i when d_i , denoted by $\widehat{y1}_{m1}$
- ③ The estimated ATET is then $\widehat{y1}_{m1} - \widehat{y0}_{m1}$

Generate data from a DGP that belongs to model ATET1 in equation (1) and compute ATET in sample from potential outcomes

```
. drop _all
. set seed 12345
. quietly do atetsim1
. describe
```

Contains data

Observations: 10,000
Variables: 9

Variable name	Storage type	Display format	Value label	Variable label
q	float	%9.0g		
x1	float	%9.0g		
x2	float	%9.0g		
e_t	float	%9.0g		
d	float	%9.0g		
y_0	float	%9.0g		
e_1	float	%9.0g		
y_1	float	%9.0g		
y	float	%9.0g		

Sorted by:

Note: Dataset has changed since last saved.

```
. generate double te = y_1 - y_0
. mean te if d==1
```

Mean estimation

Number of obs = 4,191

	Mean	Std. err.	[95% conf. interval]	
te	1.199777	.0197273	1.161101	1.238453

Use teffects to estimate ATET by RA

```
. teffects ra (y x1 x2) (d) , atet
Iteration 0: EE criterion = 1.619e-30
Iteration 1: EE criterion = 1.699e-32
Treatment-effects estimation      Number of obs      =      10,000
Estimator      : regression adjustment
Outcome model   : linear
Treatment model : none
```

y		Coefficient	Robust std. err.	z	P> z	[95% conf. interval]	
ATET	d						
(1 vs 0)		1.211919	.0231843	52.27	0.000	1.166478	1.257359
POmean	d						
0		1.761378	.0242416	72.66	0.000	1.713866	1.808891

RA for ATET by hand

RA for ATE is not consistent for this DGP

```
. sum te // estimate ATE from potential outcomes
      Variable |           Obs           Mean   Std. dev.       Min       Max
      +-----+-----+-----+-----+-----+-----+
      te       |      10,000   .3833177   1.450735  -5.269057   5.935896
. teffects ra (y x1 x2) (d) , ate
Iteration 0:   EE criterion =  1.617e-30
Iteration 1:   EE criterion =  5.597e-32
Treatment-effects estimation               Number of obs       =      10,000
Estimator      : regression adjustment
Outcome model   : linear
Treatment model : none
```

	y	Coefficient	Robust std. err.	z	P> z	[95% conf. interval]	
ATE							
	d						
	(1 vs 0)	1.204812	.0195499	61.63	0.000	1.166495	1.243129
POmean							
	d						
	0	1.233183	.0173101	71.24	0.000	1.199256	1.26711

Two-period DID

What do DID methods do?

- DID methods use a multi-period design to identify and estimate an ATET
- DID methods allow for weaker assumptions between the unobservables that affect the treatment assignment and the unobservables that affect the potential outcomes
- Just like an ATET allows for endogeneity that an ATE cannot handle, the multi-period design allows DID methods for an ATET to handle endogeneity that cross-sectional designs for the ATET cannot handle

$$\text{effect}_{did} = \underbrace{\mathbf{E}[(y_{i,1} - y_{i,0}) | d_i = 1]}_{\text{first difference over time}} - \underbrace{\mathbf{E}[(y_{i,1} - y_{i,0}) | d_i = 0]}_{\text{first difference over time}}$$

second difference over treatment status

Many designs

- There are many designs and many details to consider
- Designs vary over
 - the number of observed time periods
 - whether treatment assignment is simultaneous or staggered

Some common features

- Some common features of the designs we will consider
 - We observe data for multiple periods $t \in \{0, 1, 2, \dots, T\}$, $T \geq 1$
 - There are two treatment levels: treated and not treated (control)
 - No one is given the treatment in the first period, or in the few periods
 - Denote the first period in which some units are assigned to treatment by q
 - The treatment is irreversible
 - If a unit is assigned to the treated status, that unit remains treated for the remaining time periods
 - The parameters of interest are

$$ATE_t = \mathbf{E}[y_{1,i,t} - y_{0,i,t} | d_i = 1] \text{ for } t \geq q$$

where $y_{1,i,t}$ is the treated potential outcome of unit i at time t and $y_{0,i,t}$ is the control potential outcome of unit i at time t

Where to start?

- Before we get into general conditions and identification strategies, let's start (approximately) where the literature started
- Let's start by considering a two-period model that violates the CMI condition required for the cross-sectional ATET
- For this model
 - Let's see whether a difference-in-differences (DID) estimator could estimate the ATET of interest
 - Let's see how we might compute a DID estimator

A model that violates cross-sectional CMI

- We have two periods of panel data, $i \in \{1, 2, \dots, N\}$ and $t \in \{0, 1\}$
- No one is treated in $t = 0$, some individuals are treated at time $t = 1$
- d_i is 1 for those treated in time 1 and 0 for those not treated in time 1
- We have time-invariant covariates $\mathbf{z}_i$
- $y_{0,i,t}$ is the control potential outcome for i at time t
- $y_{1,i,t}$ is the treated potential outcome for i at time t
- $w_{i,t} = 1$ if $d_i = 1$ & $t = 1$ and is 0 otherwise

$$d_i = \mathbf{z}_i\gamma + \eta_i$$

$$y_{0,i,t} = \mathbf{z}_i\beta + \eta_i + \lambda_t + \epsilon_{0,i,t}$$

$$y_{1,i,t} = \mathbf{z}_i\beta + \eta_i + \lambda_t + w_{i,t}\delta + \epsilon_{1,i,t}$$

$$\begin{aligned}
 d_i &= \mathbf{z}_i\boldsymbol{\gamma} + \eta_i \\
 y_{0,i,t} &= \mathbf{z}_i\boldsymbol{\beta} + \eta_i + \lambda_t + \epsilon_{0,i,t} \\
 y_{1,i,t} &= \mathbf{z}_i\boldsymbol{\beta} + \eta_i + \lambda_t + w_{i,t}\delta + \epsilon_{1,i,t}
 \end{aligned}$$

- We want to know the ATET in period 1

$$\begin{aligned}
 &\mathbf{E}[y_{1,i,1} - y_{0,i,1} | d_i = 1] \\
 &= \mathbf{E}[\mathbf{z}_i\boldsymbol{\beta} | d_i = 1] + \mathbf{E}[\eta_i | d_i = 1] + \mathbf{E}[\lambda_1 | d_i = 1] + \delta + \mathbf{E}[\epsilon_{1,i,1} | d_i = 1] \\
 &\quad - \mathbf{E}[\mathbf{z}_i\boldsymbol{\beta} | d_i = 1] - \mathbf{E}[\eta_i | d_i = 1] - \mathbf{E}[\lambda_1 | d_i = 1] - \mathbf{E}[\epsilon_{0,i,1} | d_i = 1] \\
 &= \delta
 \end{aligned}$$

- The presence of η_i in the equations for d_i , $y_{0,i,t}$ and $y_{1,i,t}$ implies that we cannot use the cross-sectional ATET approach
- But a difference-in-difference approach identifies the $ATET = \delta$

$$\mathbf{E}[(y_{i,1} - y_{i,0}) | d_i = 1] - \mathbf{E}[(y_{i,1} - y_{i,0}) | d_i = 0] = \delta$$

Two-way fixed-effects estimator

The two-period two-way fixed estimator with time-invariant covariates and the time-varying treatment indicator $w_{i,t}$ can be written as

$$y_{i,t} = \delta w_{i,t} + \eta_i + \gamma 2.t_t + \epsilon_{i,t}$$

where

- $w_{i,t} = 1$ if $t = 2$ and $d_i = 1$, in other cases $w_{i,t} = 0$
- $2.t_t = 1$ if $t=2$ and 0 otherwise
- η_i is the fixed effect
- The time-invariant covariates do not appear, because they are perfectly collinear with η_i

Two-way fixed-effects estimator

$$y_{i,t} = \delta w_{i,t} + \eta_i + \gamma 2.t_t + \epsilon_{i,t} \quad (2)$$

- One way to compute a difference-in-difference estimator for two-period design is to use the two-way fixed effects estimator
- In the two-period case, the TWFE is numerically equivalent to the first-difference regression
- First differencing equation (2) and taking an expectation yields

$$\mathbf{E}[\Delta y_{i,t} | d_i] = \delta d_i + \gamma \quad (3)$$

Using equation (3) to compute a difference-in-difference yields

$$\begin{aligned} DID &= \mathbf{E}[\Delta y_{i,t} | d_i = 1] - \mathbf{E}[\Delta y_{i,t} | d_i = 0] \\ &= \delta + \gamma - \gamma \\ &= \delta \end{aligned}$$

True value of ATET is 1.5

```
. quietly do did2psim1
```

```
. describe
```

Contains data from tpsim1.dta

Observations: 20,000

Variables: 12

29 Oct 2024 15:13

Variable name	Storage type	Display format	Value label	Variable label
id	float	%9.0g		
q	float	%9.0g		
z1	float	%9.0g		
z2	float	%9.0g		
eta_i	float	%9.0g		
d	float	%9.0g		
t	float	%9.0g		
w	float	%9.0g		
lam_t	float	%9.0g		
y_0	float	%9.0g		
y_1	float	%9.0g		
y	float	%9.0g		

Sorted by: id

```
. generate double te = y_1 - y_0
```

```
. mean te if d==1 & t==2
```

Mean estimation

Number of obs = 3,961

	Mean	Std. err.	[95% conf. interval]	
te	1.534327	.022584	1.490049	1.578604

Use TWFE to estimate DID parameter

```
. xtset id t
Panel variable: id (strongly balanced)
Time variable: t, 1 to 2
Delta: 1 unit
. xtreg y i.w i.t z1 z2, fe vce(cluster id)
note: z1 omitted because of collinearity.
note: z2 omitted because of collinearity.
Fixed-effects (within) regression
Group variable: id
R-squared:
    Within = 0.2929
    Between = 0.6341
    Overall = 0.3795
Number of obs      =      20,000
Number of groups   =      10,000
Obs per group:
    min =          2
    avg =          2.0
    max =          2
F(2, 9999)         =      2074.61
Prob > F           =          0.0000
corr(u_i, Xb) = 0.4642
(Std. err. adjusted for 10,000 clusters in id)
```

y	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
1.w	1.546891	.0290333	53.28	0.000	1.48998	1.603802
2.t	-1.124304	.0182405	-61.64	0.000	-1.160059	-1.088549
z1	0	(omitted)				
z2	0	(omitted)				
_cons	2.204786	.0070956	310.73	0.000	2.190877	2.218695
sigma_u	2.1599388					
sigma_e	1.0034694					
rho	.82247884	(fraction of variance due to u_i)				

First-difference estimator

produces numerically identical results for δ as TWFE

- There are several other estimation strategies for this ATET which we need to discuss
- But first let's discuss the restrictions that we need between the unobservables that affect treatment assignment and the unobservables that affect the potential outcomes

- Consider

d_i = treatment-assignment-indicator process

$$y_{0,i,t} = \mathbf{z}_i\boldsymbol{\beta} + \eta_i + \lambda_t + \epsilon_{0,i,t} \quad (4)$$

$$y_{1,i,t} = \mathbf{z}_i\boldsymbol{\beta} + \eta_i + \lambda_t + \mathbf{w}_{i,t}\boldsymbol{\delta} + \epsilon_{1,i,t}$$

- In terms of regression models, ([Abadie, 2005](#), discussion surrounding his equation (2)) gave an exceptionally clear intuitive restriction
 - The treatment assignment process must be independent of the time-varying errors that drive the potential outcomes
 - In terms of the model in equation (4),

$$Pr(d_i = 1 | \mathbf{z}_i, \epsilon_{0,i,t}, \epsilon_{1,i,t}) = Pr(d_i = 1 | \mathbf{z}_i)$$

- In terms of the model in equation (4), the η_i are unobservables that can affect d_i , but d_i cannot depend on $\epsilon_{0,i,t}$ or $\epsilon_{1,i,t}$

DID assumptions

- The DID literature uses three key assumptions
 - ① A no-anticipation (NA) assumption
 - ② A parallel-trends (PT) assumption
 - ③ A linear in parameters (LIN) assumption, when there are covariates in the model
- The NA assumption and PT assumption rule out any dependence of the treatment-assignment process on the time-varying potential-outcome errors
- The NA assumption and the PT assumption also restrict the functional forms
- When there are covariates in the model, the LIN assumption ensures the functional forms are linear in the parameters
- Hidden in these assumptions is the requirement that the unobservable errors must be additive, so that time differencing removes the time-invariant component

NA assumption in simple two-period model

- In a two-period model with time-invariant covariates, the NA assumption is

$$\mathbf{E}[y_{1,i,0} - y_{0,i,0} | \mathbf{z}_i, d_i = 1] = 0$$

- The ATET in the period before the treatment is 0
The future treatment does not cause a response before it is given
- Treatment assignment is not related to pre-treatment differences in the potential outcomes
- Note that allowing for covariates makes this assumption more likely to hold

PT assumption in simple two-period model

- In a two-period model with time-invariant covariates, the PT assumption is

$$\mathbf{E}[y_{0,i,1} - y_{0,i,0} | \mathbf{z}_i, d_i] = \mathbf{E}[y_{0,i,1} - y_{0,i,0} | \mathbf{z}_i]$$

- The trend in the control potential outcome is not related to the treatment process, after conditioning on covariates
- After conditioning on covariates, the average of the time-difference in the control potential outcomes is the same for those who get the treatment and those who do not get the treatment

Assumptions matter

- These assumptions matter

Example using DGP without NA or PT

Additive errors assumption

- The [identification argument](#) for the two-period design reveals that there is an implicit assumption of additive errors in this DID approach
- If there errors are not additive, then differencing does not remove them
- [Wooldridge \(2021\)](#) makes this assumption explicit by adding a linear in parameters (LIN) assumption
 - This paper also [uses a constructive approach](#) to identification
 - This paper also shows the equivalence between the Mundlak approach and TWFE and uses this equivalence and it uses this equivalence formulate an equivalent cross-sectional regression with clustered standard errors [Mundlak](#)
- [Wooldridge \(2023\)](#) considers approaches for logit and exponential-mean models that are not linear in parameters and shows that using a linear model can provide a approximation that should not be used

Stata commands

- Use `xtddidregress` or `xthdidregress` for two-period panel-data designs
- `xtddidregress` uses a TWFE estimator with cluster-robust standard errors to estimate the ATET

syntax

```
xtddidregress ( ovar ocovariates ) ( tvar ) [if] [in] [weight] ,  
group(groupvar) [time(timevar) vce( cluster clustervar ) ]
```

- `xthdidregress` implements a TWFE estimator and several others estimators
 - The extra h is for heterogeneous
 - Discuss this estimator later

```
. // identical to xtreg y i.w i.t z1 z2, fe vce(cluster id)
. use tpsim1, clear
. quietly xtset id t
. xtdidregress (y z1 z2) (w) , group(id) time(t)
note: z1 omitted because of collinearity.
note: z2 omitted because of collinearity.
```

Treatment and time information

Time variable: t

Control: w = 0

Treatment: w = 1

	Control	Treatment
Group		
id	6039	3961
Time		
Minimum	1	2
Maximum	1	2

Difference-in-differences regression

Number of obs = 20,000

Data type: Longitudinal

(Std. err. adjusted for 10,000 clusters in id)

y	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
ATET						
(1 vs 0) ^w	1.546891	.0290333	53.28	0.000	1.48998	1.603802

Note: ATET estimate adjusted for covariates, panel effects, and time effects.

Repeated Cross-sectional data

- In many cases, the data are sample means from repeated cross-sectional datasets at some level of geographic or institutional level
 - Like sample mean of house prices within counties or sample mean of student test scores within schools
- This type of data should be weighted by the number of observations at the aggregation level
 - Failing to use these weights makes your standard errors too large
- Point estimates from infeasible unit-level data and the sample-mean aggregated data are the same, but the standard errors are different
 - The standard errors from the unit-level data are the correct ones
 - [Here](#) is a long, involved example and simulation

Thus far

- Covered the case of two-period panel-data designs
- DID is about estimating an ATET
- It turns out that there are several regression models that identify the ATET of interest
- DID allows time-invariant and unit-invariant unobservables to affect both the treatment and the potential outcomes
- The NA and PT assumptions restrict the dependence between the time-varying unobservables and the treatment-assignment process
- The outcome model must have additive errors

Upcoming

- Common-entry-time multiperiod designs
- A plethora of estimators
 - (Extended) TWFE
 - RA
 - IPW
 - AIPW
- simultaneous inference
- Staggered-entry-time multiperiod design

RA, IPW, and AIPW estimators

Other estimators

- There are other estimators for the ATET that we need to discuss
 - There are versions of each of these estimators for panel and for repeated cross-sectional data
- The RA estimator is the analog of the cross-sectional RA estimator
- The IPW estimator uses a model for the treatment instead of a model for the outcomes to estimate the ATET
- The AIPW estimator is an efficient combination of the RA estimator and the IPW estimator
 - The AIPW uses both an outcome model and a treatment model

- The IPW estimator becomes unstable if the predicted probability that $d_i = 1$ gets close to one
 - The overlap condition

$$Pr(d_i = 1 | \mathbf{x}_i) < 1 - \epsilon$$

rules out the predicted probability that $d_i = 1$ is one

- Near violations of this condition in the sample cause the IPW to become unstable
- RA estimator also uses this condition, but the RA is less susceptible to near violations of the overlap condition when the regression models fit well for covariate values that only appear in the other treatment status

- We have two models that we could use to construct an estimator of the ATET
 - The RA model and IPW model
- The augmented inverse probability weighted (AIPW) model and estimator is an efficient combination of the RA and the IPW
- The AIPW estimator has the double-robustness property
 - The AIPW estimator is consistent if either the outcome model or the treatment-assignment model is well specified
 - This property is remarkable, efficiency usually sacrifices robustness
- The AIPW estimator is Neyman-orthogonal, it is robust to covariate selection among the covariates and terms that enter the outcome model and the treatment-assignment model
 - Callaway, Drukker, Liu, and Sant'Anna (in progress)

History and literature

- Heckman et al. (1997) derived the RA for the two-period model
- Abadie (2005) derived the IPW for the two-period model
- Sant'Anna and Zhao (2020) derived the AIPW estimator for the two-period model
 - Sant'Anna and Zhao (2020) also provide a common framework that allows one to easily implement all three estimators
- Callaway and Sant'Anna (2021) derived versions of the RA, the IPW, and the AIPW estimators for the multiperiod model with staggered-entry times
- `xthdidregress` and `hdidregress` implement the Callaway and Sant'Anna (2021) estimators, we will discuss them later (in the section on staggered-entry designs)

Examples

```
. use rcs_sim_tp2_short, clear
. quietly didregress (y z1 z2 ) (w) [fw=fw] , group(s) time(t)
. estimates store rcsa_twfe
. quietly hdidregress ra (y z1 z2 ) (w) [fw=fw] , group(s) time(t)
. estimates store rcsa_ra
. quietly hdidregress ipw (y) (w z1 z2 ) [fw=fw] , group(s) time(t)
. estimates store rcsa_ipw
. quietly hdidregress aipw (y z1 z2) (w z1 z2 ) [fw=fw] , group(s) time(t)
. estimates store rcsa_aipw
. estimates table rcsa_twfe rcsa_ra rcsa_ipw rcsa_aipw, ///
>          b(%6.5g) se(%5.4g) modelwidth(10)                ///
>          equations(1:1:1:1) drop(Controls:)
```

Variable	rcsa_twfe	rcsa_ra	rcsa_ipw	rcsa_aipw
^w (1 vs 0)	.7873 .012			
^t 2		.7933 .02	.7884 .07	.7887 .071

Legend: b/se

More complicated DID designs

- Multiperiod common-entry time
 - In multiperiod common-entry-time designs, we observe the units for multiple periods and there is a single time period in which units are either treated or not treated
 - These designs allow the ATET to vary over time
- Multiperiod staggered-entry time
 - In multiperiod common-entry-time designs, we observe the units for multiple periods
 - Some units never get treated
 - Other units get treated at different time periods
 - The entry time is staggered, each entry time defines a cohort
 - The ATET varies over time and over cohort

More complicated DID designs

- There are versions of NA and PT that accommodate these designs
 - Mutiperiod, common entry details
 - Mutiperiod, staggered entry details
- There are versions of all the estimators (TWFE, RA, IPW, AIPW) can consistently estimate the ATETs that vary over time and cohort

Extended TWFE estimator

- The extended TWFE estimator comes from [Wooldridge \(2021\)](#)
 - Some papers note that the TWFE did not produce consistent estimates of the ATETs when the TWFE did not include terms for staggered entry, terms time-varying ATETs, and terms that allowed the effects of covariates to vary over time
 - See [Goodman-Bacon \(2021\)](#), [De Chaisemartin and d'Haultfoeuille \(2023\)](#), and [Sun and Abraham \(2021\)](#) for discussions of the problem and some partial solutions
 - [Wooldridge \(2021\)](#) showed that an extended TWFE is consistent for the ATETs of interest if one includes the extra terms to account for variations in the ATET
He derived the terms that must be included and derived several forms for the extended TWFE regressions

Callaway Sant'Anna estimators

- The RA, IPW, and AIPW estimators come from [Callaway and Sant'Anna \(2021\)](#)
 - These estimators were explicitly constructed to handle time-varying ATETs and staggered-entry times
 - They are extensions of the RA, IPW, and AIPW estimators derived by [Sant'Anna and Zhao \(2020\)](#) discussed earlier

Stata Commands

- `hdidregress` and `xthdidregress` can estimate the ATET when
 - The entry time is staggered, each entry time defines a cohort
 - The ATET varies over time and over cohort
 - `hdidregress` is for repeated cross-sectional data and `xthdidregress` is for panel data
- `hdidregress twfe` and `xthdidregress twfe` implement an extended TWFE estimator,
- `hdidregress ra` and `xthdidregress ra` implement an RA estimator,
- `hdidregress ipw` and `xthdidregress ipw` implement an IPW estimator
- `hdidregress aipw` and `xthdidregress aipw` implement an AIPW estimator

Standard errors of staggered-entry estimators

- The standard errors from the [Wooldridge \(2021\)](#) regression estimators are generally smaller than the standard errors from the [Callaway and Sant'Anna \(2021\)](#) estimators
 - Joint versus single estimation
 - The [Callaway and Sant'Anna \(2021\)](#) estimators estimate each ATET using data from the treatment period and the comparison period
 - The [Wooldridge \(2021\)](#) estimators jointly estimate all the ATETs using all the data
 - Impose versus not impose NA
 - The [Wooldridge \(2021\)](#) regression estimators impose NA by not including coefficients for those
 - The [Callaway and Sant'Anna \(2021\)](#) estimators do not impose NA and in fact estimate the pre-treatment period ATETs

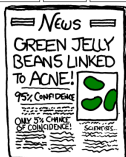
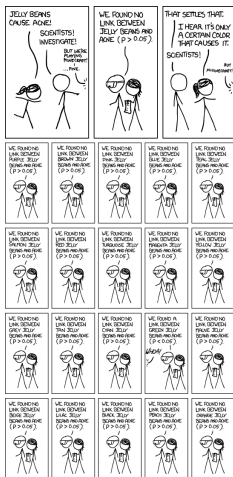
Simultaneous inference

Cohort	ATET	Robust std. err.	t	P> t	[95% conf. interval]	
t						
4	.8206642	.0346796	23.66	0.000	.7526852	.8886432
5	1.070767	.0347035	30.85	0.000	1.002742	1.138793

Note: ATET computed using covariates.

- When you have more than one parameter of interest, you must use t-tests and confidence intervals that are robust to performing multiple tests
- The t-tests and confidence intervals in standard statistical output are for single inference, not multiple inferences

Intuitive reason



More formal reasons and software

- There is a huge literature on this problem and some solutions
 - For example, see [Miller \(1966\)](#), [Hochberg and Tamhane \(1987\)](#), [Westfall and Young \(1993\)](#), and [Tamhane \(1996\)](#) for introductions
- See [Drukker \(2022\)](#) another introduction and a discussion of the user-written Stata command `sotable` that produces nominal inference using the max-t methods

```
. sotable
Max-t results
      p-value = 0.000
Critical value = 2.230
```

y		Coef.	Std. Err.	t	P> t	[95% Conf. Band]	
4	4bn.t	.8206642	.0346796	23.664	0.000	.74333	.8979984
	5.t	1.070767	.0347035	30.855	0.000	.9933799	1.148155

- Wider confidence intervals account for simultaneous inference

Take Aways

- DID methods are about estimating one or more ATETs
- DID designs allow the treatment-assignment process to depend on cluster-level unobservables that also affect the potential outcomes
- The ATETs from multi-period designs (with common entry time or staggered-entry time) can be estimated by extended TWFE, RA, IPW, or AIPW methods
- The process of
 - writing down a simple model,
 - simulating data from an example DGP, and
 - estimating things of interest from the generated datacan help illustrate how DID methods work or don't work

Bibliography

- ABADIE, A. (2005): "Semiparametric difference-in-differences estimators," *The Review of Economic Studies*, 72, 1–19.
- CALLAWAY, B. AND P. H. SANT'ANNA (2021): "Difference-in-Differences with multiple time periods," *Journal of Econometrics*, 225, 200–230.
- CAMERON, A. C. AND P. K. TRIVEDI (2022): *Microeconometrics Using Stata: Volume II*, College Station, Tx: Stata Press, second ed.
- CATTANEO, M. D., D. M. DRUKKER, AND A. D. HOLLAND (2013): "Estimation of multivalued treatment effects under conditional independence," *Stata Journal*, 13, 407–450.
- DE CHAISEMARTIN, C. AND X. D'HAULTFOEUILLE (2023): "Two-way fixed effects and differences-in-differences with heterogeneous treatment effects: A survey," *The Econometrics Journal*, 26, C1–C30.

- DRUKKER, D. M. (2022): "Simultaneous tests and confidence bands for Stata estimation commands," *The Stata Journal*, 23, 514–544.
- GOODMAN-BACON, A. (2021): "Difference-in-differences with variation in treatment timing," *Journal of Econometrics*, 225, 254–277.
- HECKMAN, J. J. (1976): "The common structure of statistical models of truncation, sample selection and limited dependent variables and a simple estimator for such models," in *Annals of Economic and Social Measurement, Volume 5, number 4*, NBER, 475–492.
- (1978): "Dummy exogenous variables in a simulation equation system," *Econometrica*, 46, 403–426.
- HECKMAN, J. J., H. ICHIMURA, AND P. TODD (1997): "Matching as an Econometric Evaluation Estimator: Evidence from Evaluating a Job Training Program," *Review of Economic Studies*, 64, 605–654.

- HOCHBERG, Y. AND A. C. TAMHANE (1987): *Multiple comparison procedures*, New York: John Wiley & Sons, Inc.
- HOLLAND, P. W. (1986): "Statistics and causal inference," *Journal of the American Statistical Association*, 945–960.
- MILLER, R. G. J. (1966): *Simultaneous statistical inference*, McGraw-Hill.
- NEWKEY, W. K. (1984): "A method of moments interpretation of sequential estimators," *Economics Letters*, 14, 201–206.
- SANT'ANNA, P. H. AND J. ZHAO (2020): "Doubly robust difference-in-differences estimators," *Journal of Econometrics*, 219, 101–122.
- SUN, L. AND S. ABRAHAM (2021): "Estimating dynamic treatment effects in event studies with heterogeneous treatment effects," *Journal of Econometrics*, 225, 175–199.
- TAMHANE, A. C. (1996): "18 Multiple comparisons," *Handbook of statistics*, 13, 587–630.

WESTFALL, P. H. AND S. S. YOUNG (1993): *Resampling-based multiple testing: Examples and methods for p-value adjustment*, John Wiley & Sons.

WOOLDRIDGE, J. M. (2010): *Econometric Analysis of Cross Section and Panel Data*, Cambridge, Massachusetts: MIT Press, second ed.

——— (2021): “Two-Way Fixed Effects, the Two-Way Mundlak Regression, and Difference-in-Differences Estimators,”
https://www.dropbox.com/sh/zj91darudf2fica/AADj_jaf5ZuS1muobgsnxS6Za?dl=0&preview=two_way_mundlak_20210928.pdf.

——— (2023): “Simple approaches to nonlinear difference-in-differences with panel data,” *The Econometrics Journal*, 26, C31–C66.

Linked appendices

Storing stuff in Macros

Local macros store numeric and character information as strings

- `local` macros are local
- There are three syntaxes for storing information in macros

```
local lname = exp
```

```
local lname "string"
```

```
local lname : extended_fcn
```

Retrieving stuff stored in macros

- Everywhere a punctuated macro name appears, its contents are substituted for the macro name.
 - The names of local macros are punctuated by enclosing them between single left quotes (‘) and single right quotes (’)

[Back to DGP](#)

Examples of local macros

```
. local value = invnorm(.975)
. display "punctuating value yields `value'"
punctuating value yields 1.959963984540054
. local vlist "y x1 x2"
. display "punctuating vlist yields `vlist'"
punctuating vlist yields y x1 x2
. local cnt : word count y x1 x2
. display "punctuating cnt yields `cnt'"
punctuating cnt yields 3
```

[Back to DGP](#)

Macro evaluation is recursive

```
. local macro1 "hello"  
. display "`macro1`"  
hello  
. local i 1  
. display "`macro`i`"  
hello
```

[Back to DGP](#)

forvalues

```
forvalues lname = #/# {  
    commands referring to 'lname'  
}
```

```
// forvalues.do  
forvalues i = 1/3 {  
    display "i is now `i'"  
}
```

```
. do forvalues  
. // forvalues.do  
. forvalues i = 1/3 {  
  2.     display "i is now `i'"  
  3. }  
i is now 1  
i is now 2  
i is now 3  
.  
end of do-file
```

forvalues II

Use forvalues when you need to change a set of variables indexed by a number

```
. drop _all
. input var1 var2 var3 var4
      var1      var2      var3      var4
1. 1 2 3 4
2. 4 5 6 6
3. end
. forvalues i=1/4 {
2.      local year = 79+`i'
3.      rename var`i' var`year'
4. }
. describe , simple
var80  var81  var82  var83
```

[Back to DGP](#)

foreach

```
foreach lname in list {  
    commands referring to 'lname'  
}
```

```
foreach lname of local lname {  
    commands referring to 'lname'  
}
```

[Back to DGP](#)

foreach II

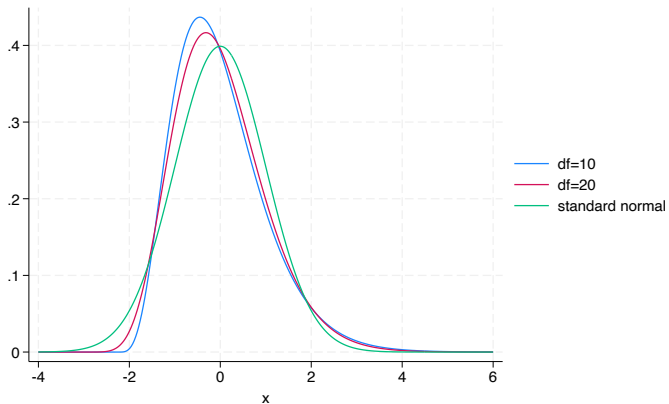
```
// foreach.do
local vlist y x1 x2
foreach v of local vlist {
    display "v is now `v'"
}
```

```
. local vlist y x1 x2
. foreach v of local vlist {
2.     display "v is now `v'"
3. }
v is now y
v is now x1
v is now x2
```

[Back to DGP](#)

Models versus DGPs

- In many examples in this talk,
 - ① we describe a model that has some property
 - ② then we draw from a DGP that has that property
- What's the difference between a model and a DGP?
- A DGP specifies everything you need to know to simulate a process on a computer
- A model is class (whole set of) DGPs
- For example, suppose that $y_i = \mathbf{x}_i\beta + \epsilon_i$
 - An example DGP specifies that $\mathbf{x}_i \sim N(0, 1)$, $\beta = 3$ and $\epsilon_i \sim N(0, 1)$
 - An example model might specify that both $\mathbf{x}_i$ and ϵ_i have finite first and second moments, that $\text{Var}(\mathbf{x}_i) \neq 0$, that $\mathbf{x}_i$ has zero correlation ϵ_i , and that β lies in a closed and bounded set of the real line



[Back to DGP](#)

Tips on simulating data

- Where possible, make the data more difficult, it will never be as difficult as the real world data
 - Correlate the covariates in the model
 - Put non-zero means in your covariates
- Sometimes the model imposes a specific distribution for an error term
 - A logit requires a standard logistic error term
- Sometimes the model only requires that the variable be IID
 - In these cases, draw from a distribution that is not normal
 - I like to use the $(\text{rchi2}(q)-q)/\sqrt{2*q}$ process with $q \in [15, 25]$

Sample size in simulations

- The larger the sample size in the simulated data, the more concentrated an estimator will be around the value to which it converges
- In practice, many DID studies use samples with few observations
- In this talk, we usually we want to compare known true values with what an estimator produces
 - Use large sample sizes
 - We could use the DGPs to simulate the behavior of estimators for smaller sample sizes

Sample-selection analogy

$$y_{0,i} = \mathbf{x}_i\beta_0 + \epsilon_{0,i}$$

$$y_{1,i} = \mathbf{x}_i\beta_1 + \epsilon_{1,i}$$

$$d_{1,i} = \begin{cases} 1 & \text{if } \mathbf{x}_i\gamma + \eta_i \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

- For those familiar with the Heckman model for sample selection,
 - the condition that unobservables that affect the treatment assignment are independent of the unobservables that affect the potential outcomes
 - is analogous to
 - the condition that unobservables that affect the indicator for being observable in the sample are independent of the unobservables that affect the outcome process
- See [Heckman \(1976\)](#), [Heckman \(1978\)](#), ([Wooldridge, 2010](#), section 19.6) or ([Cameron and Trivedi, 2022](#), section 18.9)

RA for ATE by hand, part 1

```
. regress y x1 x2 if d==0
```

Source	SS	df	MS	Number of obs = 5,743		
Model	3517.51275	2	1758.75638	F(2, 5740) = 1815.45		
Residual	5560.7575	5,740	.968773083	Prob > F = 0.0000		
				R-squared = 0.3875		
				Adj R-squared = 0.3873		
Total	9078.27025	5,742	1.5810293	Root MSE = .98426		

y	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
x1	.504075	.0162709	30.98	0.000	.4721779	.5359722
x2	.5912682	.0164074	36.04	0.000	.5591035	.6234328
_cons	1.006244	.0132329	76.04	0.000	.9803029	1.032186

```
. predict double y0_hat
(option xb assumed; fitted values)
```

```
. regress y x1 x2 if d==1
```

Source	SS	df	MS	Number of obs = 4,257		
Model	6720.63868	2	3360.31934	F(2, 4254) = 3224.93		
Residual	4432.5908	4,254	1.04198185	Prob > F = 0.0000		
				R-squared = 0.6026		
				Adj R-squared = 0.6024		
Total	11153.2295	4,256	2.62058963	Root MSE = 1.0208		

y	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
x1	.6878267	.017652	38.97	0.000	.6532196	.7224338
x2	.8121768	.017583	46.19	0.000	.777705	.8466486
_cons	1.311057	.0201942	64.92	0.000	1.271466	1.350648

```
. predict double y1_hat
(option xb assumed; fitted values)
```

RA for ATE by hand, part 2

```
. generate double te_hat = y1_hat - y0_hat  
. mean te_hat
```

Mean estimation

Number of obs = 10,000

	Mean	Std. err.	[95% conf. interval]	
te_hat	.390398	.0035225	.3834932	.3973028

- Standard errors from mean are wrong, because mean does not know that te_hat was created from estimated parameters
- Theoretical answer is to stack the moment conditions from the OLS regressions and the sample mean
 - see Newey (1984), <https://blog.stata.com/2014/12/08/using-gmm-to-solve-two-step-estimation-problems/>, <https://blog.stata.com/2016/09/13/an-ordered-probit-inverse-probability-weighted-ipw-es> and Methods and formulas in teffects aipw in [Stata Causal Inference and Treatment-Effects Estimation](#),
- Practical (Stata) answer is to use margins

[Back to RA by teffects](#)

RA for ATE by hand, using margins

```
. regress y i.d i.d#c.(x1 x2), vce(robust)
```

Linear regression

```
Number of obs      =      10,000
F(5, 9994)         =     3107.24
Prob > F           =      0.0000
R-squared          =      0.6096
Root MSE          =      .99997
```

y	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
1.d	.3048128	.0241354	12.63	0.000	.2575025	.3521231
d#c.x1						
0	.504075	.0167426	30.11	0.000	.4712561	.536894
1	.6878267	.0175669	39.15	0.000	.653392	.7222614
d#c.x2						
0	.5912682	.0163811	36.09	0.000	.559158	.6233783
1	.8121768	.017108	47.47	0.000	.7786416	.845712
_cons	1.006244	.0132468	75.96	0.000	.980278	1.032211

```
. margins r.d, vce(unconditional) contrast(nowald)
```

Contrasts of predictive margins

Number of obs = 10,000

Expression: Linear prediction, predict()

	Contrast	Unconditional std. err.	[95% conf. interval]	
d (1 vs 0)	.390398	.0233807	.344567	.4362289

ATET identification

- The CMI for the ATET only requires that $\mathbf{E}[y_{0,i}|\mathbf{x}_i, d_i = 0] = \mathbf{E}[y_{0,i}|\mathbf{x}_i, d_i = 1]$ which implies that $\mathbf{E}[y_i|\mathbf{x}_i, d_i = 0] = \mathbf{E}[y_{0,i}|\mathbf{x}_i]$
- ATET identification from outcome

$$\begin{aligned} ATET &= \mathbf{E}[y_{1,i} - y_{0,i} | d_i = 1] \\ &= \mathbf{E}[y_{1,i} | d_i = 1] - \mathbf{E}[y_{0,i} | d_i = 1] \\ &= \mathbf{E}[y_i | d_i = 1] - \mathbf{E}[y_{0,i} | d_i = 1] && (y_i = y_{1,i} \text{ when } d_i = 1) \\ &= \mathbf{E}[y_i | d_i = 1] - \mathbf{E}[\mathbf{E}[y_{0,i} | \mathbf{x}_i, d_i = 1] | d_i = 1] && \text{Iter E} \\ &= \mathbf{E}[y_i | d_i = 1] - \mathbf{E}[\mathbf{E}[y_{0,i} | \mathbf{x}_i, d_i = 0] | d_i = 1] && \text{by CMI} \\ &= \mathbf{E}[y_i | d_i = 1] - \mathbf{E}[\mathbf{E}[y_i | \mathbf{x}_i, d_i = 0] | d_i = 1] && \text{observed} \end{aligned}$$

The DGP implemented in this do-file satisfies ATET CMI but not ATE CMI

Listing 2: atetsim1.do

```
1 // ATET DGP correlates errors of d and y_1
2 clear all
3 local n = 10000
4 local df = 25
5
6 local dist " (rchi2(`df') - `df')/sqrt(2*`df') "
7
8 set obs `n'
9 generate q = `dist'
10 generate x1 = (q + `dist')/sqrt(2) + .2
11 generate x2 = (q + `dist')/sqrt(2) + .2
12
13 generate e_t = rlogistic() // var(e_t)=_pi^2/3
14 generate d = -0.7 + .6*x1 + .8*x2 + e_t > 0
15
16 generate y_0 = 1 + .5*x1 + .6*x2 + `dist'
17 generate e_1 = (`dist' + e_t)/sqrt(1 + _pi^2/3)
18 generate y_1 = 1.3 + .7*x1 + .8*x2 + e_1
19 generate y = d*y_1 + (1-d)*y_0
```

RA for ATET by hand, using margins

```
. regress y x1 x2 if d==0
```

Source	SS	df	MS	Number of obs	=	5,809
Model	3666.85513	2	1833.42756	F(2, 5806)	=	1848.94
Residual	5757.29994	5,806	.991612115	Prob > F	=	0.0000
				R-squared	=	0.3891
				Adj R-squared	=	0.3889
Total	9424.15507	5,808	1.62261623	Root MSE	=	.9958

y	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
x1	.5118037	.0165688	30.89	0.000	.4793225	.5442848
x2	.5860769	.0163157	35.92	0.000	.554092	.6180617
_cons	1.001102	.0132933	75.31	0.000	.9750426	1.027162

```
. predict double y0_hat, xb
```

```
. sum y0_hat if d==1
```

Variable	Obs	Mean	Std. dev.	Min	Max
y0_hat	4,191	1.761378	.9056168	-1.09787	5.362821

```
. local y0t = r(mean)
```

```
. sum y if d==1
```

Variable	Obs	Mean	Std. dev.	Min	Max
y	4,191	2.973297	1.205172	.0522079	8.219422

```
. local y1t = r(mean)
```

```
. di "estimated atet is " `y1t' - `y0t'
estimated atet is 1.2119187
```

RA for ATET by teffects

$$d_i = \mathbf{z}_i\gamma + \eta_i$$

$$y_{0,i,t} = \mathbf{z}_i\beta + \eta_i + \lambda_t + \epsilon_{0,i,t} \quad \mathbf{E}[\epsilon_{1,i,1}|d = 1] = \mathbf{E}[\epsilon_{1,i,0}|d = 1]$$

$$y_{1,i,t} = \mathbf{z}_i\beta + \eta_i + \lambda_t + w_{i,t}\delta + \epsilon_{1,i,t} \quad \mathbf{E}[\epsilon_{0,i,1}|d = 0] = \mathbf{E}[\epsilon_{0,i,0}|d = 0]$$

$$\mathbf{E}[y_{i,t}|d_i] = \mathbf{E}[y_{d_i,i,t}|d_i]$$

$$\begin{aligned} \theta_{did} &= \mathbf{E}[(y_{i,1} - y_{i,0})|d_i = 1] - \mathbf{E}[(y_{i,1} - y_{i,0})|d_i = 0] \\ &= \mathbf{E}[y_{i,1}|d_i = 1] - \mathbf{E}[y_{i,0}|d_i = 1] - (\mathbf{E}[y_{i,1}|d_i = 0] - \mathbf{E}[y_{i,0}|d_i = 0]) \\ &= \mathbf{E}[y_{1,i,1}|d_i = 1] - \mathbf{E}[y_{1,i,0}|d_i = 1] \\ &\quad - (\mathbf{E}[y_{0,i,1}|d_i = 0] - \mathbf{E}[y_{0,i,0}|d_i = 0]) \\ &= \mathbf{E}[\mathbf{z}_i|d_i = 1]\beta + \mathbf{E}[\eta_i|d_i = 1] + \mathbf{E}[\lambda_1|d_i = 1] + \delta + \mathbf{E}[\epsilon_{1,i,1}|d_i = 1] \\ &\quad - (\mathbf{E}[\mathbf{z}_i|d_i = 1]\beta + \mathbf{E}[\eta_i|d_i = 1] + \mathbf{E}[\lambda_0|d_i = 1] + \mathbf{E}[\epsilon_{1,i,0}|d_i = 1]) \\ &\quad - (\mathbf{E}[\mathbf{z}_i|d_i = 0]\beta + \mathbf{E}[\eta_i|d_i = 0] + \mathbf{E}[\lambda_1|d_i = 0] + \mathbf{E}[\epsilon_{0,i,1}|d_i = 0]) \\ &\quad + (\mathbf{E}[\mathbf{z}_i|d_i = 0]\beta + \mathbf{E}[\eta_i|d_i = 0] + \mathbf{E}[\lambda_0|d_i = 0] + \mathbf{E}[\epsilon_{0,i,0}|d_i = 0]) \\ &= \delta \end{aligned}$$

The DGP implemented in this do-file satisfies DID but not cross-sectional ATET CMI

Listing 3: did2psim1.do

```
1  clear all
2  local n = 10000
3  local df = 25
4  local dist " (rchi2(`df') - `df')/sqrt(2*`df') "
5  set obs `n'
6  generate id = _n
7  generate q = `dist'
8  generate z1 = (q + `dist')/sqrt(2) + .2
9  generate z2 = (q + `dist')/sqrt(2) + .2
10 generate eta_i = rlogistic()
11 generate d = -0.7 + .6*z1 + .5*z2 + eta_i > 0
12 expand 2
13 sort id
14 by id: generate t = _n
15 local lam_1 = `dist'
16 local lam_2 = `dist'
17 generate w = (d==1) & (t==2)
18 generate lam_t = cond(t==1, `lam_1', `lam_2')
19 generate y_0 = 1 + .5*z1 + .6*z2 + eta_i + lam_t + `dist'
20 generate y_1 = 1 + .5*z1 + .6*z2 + eta_i + lam_t + `dist' + w*1.5
21 generate y = d*y_1 + (1-d)*y_0
22 save tpsim1, replace
```

Use first-difference regression to estimate DID parameter

```
. regress d.y d, vce(cluster id)
```

Linear regression

```
Number of obs   =    10,000
F(1, 9999)      =   2838.75
Prob > F        =    0.0000
R-squared       =    0.2214
Root MSE       =    1.4191
```

(Std. err. adjusted for 10,000 clusters in id)

D.y	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
d	1.546891	.0290333	53.28	0.000	1.48998	1.603802
_cons	-1.124304	.0182405	-61.64	0.000	-1.160059	-1.088549

[Back to simple TWFE](#)

Aside on strict exogeneity

d_i = treatment-assignment-indicator process

$$y_{0,i,t} = \mathbf{z}_i\boldsymbol{\beta} + \eta_i + \lambda_t + \epsilon_{0,i,t}$$

$$y_{1,i,t} = \mathbf{z}_i\boldsymbol{\beta} + \eta_i + \lambda_t + w_{i,t}\delta + \epsilon_{1,i,t}$$

- For those familiar with panel-data models ([Wooldridge, 2010](#), page 301), the restriction that d_i cannot depend on the potential-outcome errors in any period is a type of strict-exogeneity assumption

Back to DID exogeneity

- The conditional on covariates versions of NA and PT are more believable and allow for more flexible models in practice
- The unconditional versions of NA and PT can be used to simplify some derivations
- Unconditional, two-period NA

$$\mathbf{E}[y_{1,i,0} - y_{0,i,0} | d_i = 1] = 0$$

- Unconditional two-period PT

$$\mathbf{E}[y_{0,i,1} - y_{0,i,0} | d_i] = \mathbf{E}[y_{0,i,1} - y_{0,i,0}]$$

which implies

$$\mathbf{E}[y_{0,i,1} - y_{0,i,0} | d_i = 1] = \mathbf{E}[y_{0,i,1} - y_{0,i,0} | d_i = 0]$$

- DID yields ATET under unconditional NA & unconditional PT in two-period model

$$\begin{aligned}
& \mathbf{E}[(y_{i,1} - y_{i,0})|d_i = 1] - \mathbf{E}[(y_{i,1} - y_{i,0})|d_i = 0] \\
&= \mathbf{E}[y_{i,1}|d_i = 1] - \mathbf{E}[y_{i,0}|d_i = 1] - \mathbf{E}[y_{i,1}|d_i = 0] + \mathbf{E}[y_{i,0}|d_i = 0] \\
&= \mathbf{E}[y_{1,i,1}|d_i = 1] - \mathbf{E}[y_{1,i,0}|d_i = 1] - \mathbf{E}[y_{0,i,1}|d_i = 0] + \mathbf{E}[y_{0,i,0}|d_i = 0] \\
&= \mathbf{E}[y_{1,i,1}|d_i = 1] - \mathbf{E}[y_{0,i,1}|d_i = 1] \\
&\quad + \mathbf{E}[y_{0,i,1}|d_i = 1] - \mathbf{E}[y_{1,i,0}|d_i = 1] - \mathbf{E}[y_{0,i,1}|d_i = 0] + \mathbf{E}[y_{0,i,0}|d_i = 0] \\
&= ATET + \mathbf{E}[y_{0,i,1}|d_i = 1] - \mathbf{E}[y_{1,i,0}|d_i = 1] \\
&\quad - \mathbf{E}[y_{0,i,1}|d_i = 0] + \mathbf{E}[y_{0,i,0}|d_i = 0] \\
&= ATET + \mathbf{E}[y_{0,i,1}|d_i = 1] - \mathbf{E}[y_{0,i,0}|d_i = 1] \\
&\quad - \mathbf{E}[y_{0,i,1}|d_i = 0] + \mathbf{E}[y_{0,i,0}|d_i = 0] \text{ by NA} \\
&= ATET + \mathbf{E}[(y_{0,i,1} - y_{0,i,0})|d_i = 1] \\
&\quad - \left(\mathbf{E}[(y_{0,i,1} - y_{0,i,0})|d_i = 0] \right) \\
&= ATET \text{ by PT}
\end{aligned}$$

- Wooldridge (2021) uses a constructive approach to show how the NA, PT, and LIN assumptions imply that a DID framework identifies an ATET of interest
 - Wooldridge (2021) constructs the class of observable outcomes that could be generated from potential outcomes that satisfy NA, PT, and LIN
 - Plugging this constructed observed outcome into a DID expression yields the ATET of interest

[Back to additive errors](#)

- For example, in a two-period model without covariates the means of the observable outcomes given treatment d_i that could be generated from potential outcomes that satisfy the NA and PT assumption are

$$\mathbf{E}[y_{i,0}|d_i] = \lambda + \alpha d_i + \theta_0$$

$$\mathbf{E}[y_{i,1}|d_i] = \lambda + \alpha d_i + \theta_1 + d_i \tau_1$$

where τ_1 is the ATET of interest

$$\begin{aligned} & \mathbf{E}[y_{i,1}|d_i = 1] - \mathbf{E}[y_{i,0}|d_i = 1] - \{\mathbf{E}[y_{i,1}|d_i = 0] - \mathbf{E}[y_{i,0}|d_i = 0]\} \\ &= \lambda + \alpha + \theta_1 + \tau_1 - (\lambda + \alpha + \theta_0) - [\lambda + \theta_1 - (\lambda + \theta_0)] \\ &= \theta_1 - \theta_0 + \tau_1 - (\theta_1 - \theta_0) \\ &= \tau_1 \end{aligned}$$

- Wooldridge (2021) uses the equivalence between the TWFE and the two-way Mundlak (TWM) regression to show that the coefficient on w in the regression
`regress y w d 2.t`
is numerically identical to the coefficient in the TWFE regression
`xtreg y w 2.t, fe`

[Back to additive errors](#)

Illustrate regression equivalence

[Back to additive errors](#)

```
. regress y w d 2.t, vce(cluster id)
```

```
Linear regression
```

Number of obs	=	20,000
F(3, 9999)	=	7262.54
Prob > F	=	0.0000
R-squared	=	0.5979
Root MSE	=	1.6427

(Std. err. adjusted for 10,000 clusters in id)

y	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
w	1.546891	.029034	53.28	0.000	1.489978	1.603803
d	3.213639	.0330719	97.17	0.000	3.148811	3.278467
2.t	-1.124304	.018241	-61.64	0.000	-1.16006	-1.088548
_cons	.9318637	.0215062	43.33	0.000	.8897072	.9740201

```
. xtreg y i.w i.t , fe vce(cluster id)
```

```
Fixed-effects (within) regression
```

Number of obs	=	20,000
Number of groups	=	10,000

```
Group variable: id
```

R-squared:

Within	=	0.2929	min	=	2
Between	=	0.6341	avg	=	2.0
Overall	=	0.3795	max	=	2

F(2, 9999)	=	2074.61
Prob > F	=	0.0000

```
corr(u_i, Xb) = 0.4642
```

(Std. err. adjusted for 10,000 clusters in id)

y	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
1.w	1.546891	.0290333	53.28	0.000	1.48998	1.603802
2.t	-1.124304	.0182405	-61.64	0.000	-1.160059	-1.088549
_cons	2.204786	.0070956	310.73	0.000	2.190877	2.218695

$\epsilon_{0,i,t}$ and $\epsilon_{1,i,t}$ are mean zero and are independent of $\mathbf{z}_i$ and of η_i

$$t \in \{1, 2\}$$

$$d_i = \mathbf{z}_i \gamma + (\eta_i + \epsilon_{0,i,1}) / \text{sqrt}(2)$$

$$y_{0,i,t} = \mathbf{z}_i \beta + \eta_i + \lambda_t + \epsilon_{0,i,t}$$

$$y_{1,i,t} = \mathbf{z}_i \beta + \eta_i + \lambda_t + w_{i,t} \delta + \epsilon_{1,i,t}$$

- The above model violates both NA and PT
- NA requires that

$$\mathbf{E}[y_{1,i,0} - y_{0,i,0} | \mathbf{z}_i, d_i = 1] = 0$$

But

$$\begin{aligned} \mathbf{E}[y_{1,i,0} - y_{0,i,0} | \mathbf{z}_i, d_i = 1] &= \mathbf{E}[\epsilon_{1,i,0} - \epsilon_{0,i,0} | \mathbf{z}_i, d_i = 1] \\ &= \mathbf{E}[\epsilon_{1,i,0} | \mathbf{z}_i, d_i = 1] - \mathbf{E}[\epsilon_{0,i,0} | \mathbf{z}_i, d_i = 1] \\ &= -\mathbf{E}[\epsilon_{0,i,0} | \mathbf{z}_i, d_i = 1] \\ &\neq 0 \text{ because } \epsilon_{0,i,0} \text{ is in eq for } d_i \end{aligned}$$

$\epsilon_{0,i,t}$ and $\epsilon_{1,i,t}$ are mean zero and are independent of $\mathbf{z}_i$ and of η_i

$$t \in \{1, 2\}$$

$$d_i = \mathbf{z}_i \boldsymbol{\gamma} + (\eta_i + \epsilon_{0,i,1}) / \text{sqrt}(2)$$

$$y_{0,i,t} = \mathbf{z}_i \boldsymbol{\beta} + \eta_i + \lambda_t + \epsilon_{0,i,t}$$

$$y_{1,i,t} = \mathbf{z}_i \boldsymbol{\beta} + \eta_i + \lambda_t + w_{i,t} \delta + \epsilon_{1,i,t}$$

- PT requires that

$$\mathbf{E}[y_{0,i,2} - y_{0,i,1} | \mathbf{z}_i, d_i = 1] - \mathbf{E}[y_{0,i,2} - y_{0,i,1} | \mathbf{z}_i, d_i = 0] = 0$$

But

$$\begin{aligned} & \mathbf{E}[y_{0,i,2} - y_{0,i,1} | \mathbf{z}_i, d_i = 1] - \mathbf{E}[y_{0,i,2} - y_{0,i,1} | \mathbf{z}_i, d_i = 0] \\ &= \mathbf{E}[\lambda_2 - \lambda_1 + \epsilon_{0,i,2} - \epsilon_{0,i,1} | \mathbf{z}_i, d_i = 1] \\ & \quad - \mathbf{E}[\lambda_2 - \lambda_1 + \epsilon_{0,i,2} - \epsilon_{0,i,1} | \mathbf{z}_i, d_i = 0] \\ &= \mathbf{E}[\lambda_2 - \lambda_1 | \mathbf{z}_i, d_i = 1] - \mathbf{E}[\lambda_2 - \lambda_1 | \mathbf{z}_i, d_i = 0] \\ & \quad + \mathbf{E}[\epsilon_{0,i,2} - \epsilon_{0,i,1} | \mathbf{z}_i, d_i = 1] + \mathbf{E}[\epsilon_{0,i,2} - \epsilon_{0,i,1} | \mathbf{z}_i, d_i = 0] \\ &= -\mathbf{E}[\epsilon_{0,i,1} | \mathbf{z}_i, d_i = 1] - \mathbf{E}[\epsilon_{0,i,1} | \mathbf{z}_i, d_i = 0] \\ &\neq 0 \text{ because } \epsilon_{0,i,1} \text{ is in eq for } d_i \end{aligned}$$

Listing 4: did2psim2.do: time-invariant covariates

```
1 // violates NA and PT assumptions e01 enters treatment
2 clear all
3 local n = 10000
4 local df = 25
5 local dist " (rchi2(`df') - `df')/sqrt(2*`df') "
6 set obs `n'
7 generate id = _n
8 generate q = `dist'
9 generate z1 = (q + `dist')/sqrt(2) + .2
10 generate z2 = (q + `dist')/sqrt(2) + .2
11 drawnorm eta_i e01 e02
12 generate d = -0.7 + .6*z1 + .5*z2 + (eta_i + e01)/sqrt(2) > 0
13 expand 2
14 sort id
15 by id: generate t = _n
16 local lam_1 = `dist'
17 local lam_2 = `dist'
18 generate w = (d==1) & (t==2)
19 generate lam_t = cond(t==1, `lam_1', `lam_2')
20 generate e0 = cond(t==1, e01, e02)
21 generate e1 = `dist'
22 generate y_0 = 1 + .5*z1 + .6*z2 + eta_i + lam_t + e0
23 generate y_1 = 1 + .5*z1 + .6*z2 + eta_i + lam_t + e1 + w*1.5
24 generate y = d*y_1 + (1-d)*y_0
25 save tpsim2, replace
```

Get data and estimate ATET from potential outcomes

```
. drop _all  
. set seed 12345  
. quietly do did2psim2  
. describe
```

Contains data from tpsim2.dta

Observations: 20,000

Variables: 16

29 Oct 2024 15:13

Variable name	Storage type	Display format	Value label	Variable label
id	float	%9.0g		
q	float	%9.0g		
z1	float	%9.0g		
z2	float	%9.0g		
eta_i	float	%9.0g		
e01	float	%9.0g		
e02	float	%9.0g		
d	float	%9.0g		
t	float	%9.0g		
w	float	%9.0g		
lam_t	float	%9.0g		
e0	float	%9.0g		
e1	float	%9.0g		
y_0	float	%9.0g		
y_1	float	%9.0g		
y	float	%9.0g		

Sorted by: id

[Back to assumptions matter](#)

Estimate ATET from potential outcomes and sample means of e1 e0 over d and t

```
. generate double te = y_1 - y_0
```

```
. mean te if d==1 & t==2
```

Mean estimation

Number of obs = 3,622

	Mean	Std. err.	[95% conf. interval]	
te	1.520087	.023647	1.473724	1.56645

```
. qui mean e0 e1 , over(d t)
```

```
. sutable
```

Max-t results

p-value = 0.000

Critical value = 2.724

Mean	Coef.	Std. Err.	t	P> t	[95% Conf. Band]	
c.e0@0bn.d#1bn.t	-.3063405	.0116354	-26.328	0.000	-.3380383	-.2746428
c.e0@0bn.d#2.t	-.0216851	.0124646	-1.740	0.495	-.0556417	.0122715
c.e0@1.d#1bn.t	.5197236	.0151335	34.343	0.000	.4784963	.560951
c.e0@1.d#2.t	-.0194364	.0166912	-1.164	0.893	-.0649074	.0260347
c.e1@0bn.d#1bn.t	-.0029029	.0123385	-0.235	1.000	-.0365161	.0307102
c.e1@0bn.d#2.t	.0025503	.0125713	0.203	1.000	-.031697	.0367976
c.e1@1.d#1bn.t	-.0247736	.0168202	-1.473	0.702	-.0705959	.0210487
c.e1@1.d#2.t	.0006504	.0167816	0.039	1.000	-.0450669	.0463676

[Back to assumptions matter](#)

First-difference regression estimator is not consistent

```
. xtset id t
Panel variable: id (strongly balanced)
Time variable: t, 1 to 2
Delta: 1 unit
```

```
. regress d.y d, vce(cluster id)
```

```
Linear regression               Number of obs   =      10,000
                                F(1, 9999)       =      1785.39
                                Prob > F         =      0.0000
                                R-squared        =      0.1548
                                Root MSE     =      1.3936

                                (Std. err. adjusted for 10,000 clusters in id)
```

D.y	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
d	1.240769	.0293646	42.25	0.000	1.183208	1.298329
_cons	-1.145417	.0171525	-66.78	0.000	-1.179039	-1.111794

[Back to assumptions matter](#)

Repeated cross-section designs

- In many cases, we don't have panel data
- We observe cluster-level means from different cross-sections
- Thinking about how this data could be generated yields an insight
 - ① We might want to include frequency weights when we observe cluster-level means for each cross section
 - ② The current estimation frameworks require that the covariates be time-invariant

There are reasonable models that allow the covariates to change over time

See [Wooldridge \(2021\)](#), [Sant'Anna and Zhao \(2020\)](#) and [Callaway and Sant'Anna \(2021\)](#)

An infeasible model

- To see these points, consider the following infeasible model
- Suppose that we observed the individual-level data from repeated cross-sections of units within clusters
- The treatment assignment is at the cluster level, as is frequently the case
- For example
 - We have two cross-sectional samples ($t \in \{1, 2\}$) of individual students i within schools s
 - We see the same schools in periods 1 and 2
 - But the students are different between periods 1 and 2
 - In period 1, no schools are treated
 - In period 2, some of the schools are treated

$$\begin{aligned}
y_{s,i,1} &= d_s y_{1,s,i,1} + (1 - d_s) y_{0,s,i,1} \\
y_{s,i',2} &= d_s y_{1,s,i',2} + (1 - d_s) y_{0,s,i',2} \\
y_{0,s,i,1} &= \mathbf{z}_{s,i,1} \gamma + \alpha_s + \lambda_1 + \epsilon_{0,s,i,1} \\
y_{1,s,i,1} &= \mathbf{z}_{s,i,1} \gamma + \alpha_s + \lambda_1 + \epsilon_{1,s,i,1} \\
y_{0,s,i',2} &= \mathbf{z}_{s,i',2} \gamma + \alpha_s + \lambda_2 + \epsilon_{0,s,i',2} \\
y_{1,s,i',2} &= \delta + \mathbf{z}_{s,i',2} \gamma + \alpha_s + \lambda_2 + \epsilon_{1,s,i',2} \\
d_s &= (\bar{\mathbf{z}}_{s,1} \beta_z + \mathbf{x}_s \beta_x + \alpha_s > 0)
\end{aligned}$$

- We use i to index the students in cross-section 1 and i' to index the students in cross-section 2
- $\mathbf{z}_{s,i,1}$ are covariates of unit i in school s in sample 1 $\mathbf{z}_{s,i',2}$ are covariates of unit i' in school s in sample 2
 - Even if these characteristics are time-invariant for each student i , the characteristics vary over time
- $\bar{\mathbf{z}}_{s,1}$ are within s sample means of the covariates using the cross-sectional sample when $t = 1$
- $\mathbf{x}_s$ are school level covariates

$$\begin{aligned}
y_{s,i,1} &= d_s y_{1,s,i,1} + (1 - d_s) y_{0,s,i,1} \\
y_{s,i',2} &= d_s y_{1,s,i',2} + (1 - d_s) y_{0,s,i',2} \\
y_{0,s,i,1} &= \mathbf{z}_{s,i,1} \boldsymbol{\gamma} + \alpha_s + \lambda_1 + \epsilon_{0,s,i,1} \\
y_{1,s,i,1} &= \mathbf{z}_{s,i,1} \boldsymbol{\gamma} + \alpha_s + \lambda_1 + \epsilon_{1,s,i,1} \\
y_{0,s,i',2} &= \mathbf{z}_{s,i',2} \boldsymbol{\gamma} + \alpha_s + \lambda_2 + \epsilon_{0,s,i',2} \\
y_{1,s,i',2} &= \delta + \mathbf{z}_{s,i',2} \boldsymbol{\gamma} + \alpha_s + \lambda_2 + \epsilon_{1,s,i',2} \\
d_s &= (\bar{\mathbf{z}}_{s,1} \boldsymbol{\beta}_z + \mathbf{x}_s \boldsymbol{\beta}_x + \alpha_s > 0)
\end{aligned}$$

- $\mathbf{z}_{s,i,1}$ and $\mathbf{z}_{s,i',2}$ cannot be related to $\epsilon_{0,s,i,1}$, $\epsilon_{0,s,i',2}$, $\epsilon_{1,s,i,1}$, or $\epsilon_{1,s,i',2}$
- $\mathbf{z}_{s,i,1}$ and $\mathbf{z}_{s,i',2}$ could depend on α_s
- Allowing $\mathbf{z}_{s,i,1}$ and $\mathbf{z}_{s,i',2}$ to depend on λ_t is possible, but then the covariates becoming time-varying

Listing 5: rcs_sim1.do

```

1  // repeated cross-sectional data, i within s foreach t=1,2,
2  // treatment-assignment at level s
3  // collapse down to s and t data
4  local n1 = 10           // # of students in each school period 1
5  local n2 = 15           // # of students in each school period 2
6  local S = 5000
7  local df = 25
8  local dist " (rchi2(`df') - `df')/sqrt(2*`df') "
9  set obs `S'
10 generate s=_n
11 generate alpha_s = rlogistic()
12 generate x1      = `dist' + .2
13 generate x2      = `dist' + .2
14 expand 2
15 sort s
16 by s: generate t=_n
17 generate n = cond(t==1, `n1', `n2')
18 expand n
19 sort t s
20 by t s: generate id = (t-1)*100 + _n // id's differ by sample
21 generate z1      = (alpha_s + `dist')/sqrt(2) + .2
22 generate z2      = (alpha_s + `dist')/sqrt(2) + .2
23 local lam_1 = `dist'
24 local lam_2 = `dist'
25 generate lam_t = cond(t==1, `lam_1', `lam_2')
26 generate d      = (.1*x1 + .1*x2 + - 0.15 + alpha_s > 0)
27 generate w = d==1 & t==2
28 generate y1 = w*.8 + 1 + .3*z1 + .5*z2 + alpha_s + lam_t + `dist'
29 generate y0 = 1 + .3*z1 + .5*z2 + alpha_s + lam_t + `dist'
30 generate y = d*y1 + (1-d)*y0
31 collapse y y0 y1 d w z1 z2 x1 x2 , by(s t)
32 generate fw = cond(t==1, `n1', `n2')
33 save rcs_sim1, replace

```

Draw and describe simulated data

```
. drop _all  
. set seed 12345  
. quietly do rcs_sim1  
. describe
```

Contains data from rcs_sim1.dta

Observations: 10,000

Variables: 12

29 Oct 2024 15:13

Variable name	Storage type	Display format	Value label	Variable label
s	float	%9.0g		
t	float	%9.0g		
y	float	%9.0g	(mean) y	
y0	float	%9.0g	(mean) y0	
y1	float	%9.0g	(mean) y1	
d	float	%9.0g	(mean) d	
w	float	%9.0g	(mean) w	
z1	float	%9.0g	(mean) z1	
z2	float	%9.0g	(mean) z2	
x1	float	%9.0g	(mean) x1	
x2	float	%9.0g	(mean) x2	
fw	float	%9.0g		

Sorted by: s t

[Back to RCS quick](#)

Look at distribution of d and sample ATET from potential outcomes

```
. // treatment is constant within school
```

```
. xtset s
```

Panel variable: s (balanced)

```
. xtsum d
```

Variable		Mean	Std. dev.	Min	Max	Observations
d	overall	.4676	.4989741	0	1	N = 10000
	between		.498999	0	1	n = 5000
	within		0	.4676	.4676	T = 2

```
. if r(sd_w)>1e-14 {  
  . di "treatment varies within s"  
  . exit 498  
  . }
```

```
. tab d if t==1
```

(mean) d	Freq.	Percent	Cum.
0	2,662	53.24	53.24
1	2,338	46.76	100.00
Total	5,000	100.00	

```
. generate te = y1 - y0  
. mean te if d==1, over(t)
```

Mean estimation

Number of obs = 4,676

	Mean	Std. err.	[95% conf. interval]	
c.te@t				
1	-.0108046	.009213	-.0288664	.0072573
2	.797557	.0074978	.7828579	.8122562

Estimate ATET using xtreg

Note that z covariates time-varying, and thus not dropped

```
. xtset s
Panel variable: s (balanced)
. xtreg y i.w i.t z1 z2 , fe vce(cluster s)
Fixed-effects (within) regression
Group variable: s
R-squared:
    Within = 0.9506
    Between = 0.9897
    Overall = 0.8856
Number of obs   =    10,000
Number of groups =     5,000
Obs per group:
    min =    2
    avg  =    2.0
    max  =    2
F(4, 4999)      =   23780.84
Prob > F        =    0.0000
corr(u_i, Xb) = 0.8035
(Std. err. adjusted for 5,000 clusters in s)
```

y	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
1.w	.8136551	.0115242	70.60	0.000	.7910627	.8362475
2.t	-2.108459	.008024	-262.77	0.000	-2.12419	-2.092729
z1	.3247165	.0194777	16.67	0.000	.2865316	.3629013
z2	.5307775	.0198578	26.73	0.000	.4918476	.5697074
_cons	1.956461	.006159	317.66	0.000	1.944386	1.968535
sigma_u	1.7479009					
sigma_e	.28816675					
rho	.97353893	(fraction of variance due to u_i)				

Estimate ATET using didregress

Point estimate is identical, but standard error is much larger

```
. didregress (y z1 z2 ) (w) , group(s) time(t)
```

Treatment and time information

Time variable: t

Control: w = 0

Treatment: w = 1

	Control	Treatment
Group		
s	2662	2338
Time		
Minimum	1	2
Maximum	1	2

Difference-in-differences regression

Number of obs = 10,000

Data type: Repeated cross-sectional

(Std. err. adjusted for 5,000 clusters in s)

y	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]
ATET					
(1 vs 0) w	.8136551	.0163001	49.92	0.000	.7816998 .8456104

Note: ATET estimate adjusted for covariates, group effects, and time effects.

[Back to RCS quick](#)

Estimate ATET using didregress

Using frequency weights fixes standard error

```
. didregress (y z1 z2 ) (w) [fw=fw] , group(s) time(t)
```

Treatment and time information

Time variable: t

Control: w = 0

Treatment: w = 1

	Control	Treatment
Group		
s	2662	2338
Time		
Minimum	1	2
Maximum	1	2

Difference-in-differences regression

Number of obs = 125,000

Data type: Repeated cross-sectional

(Std. err. adjusted for 5,000 clusters in s)

y	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
ATET						
w						
(1 vs 0)	.8136551	.0117596	69.19	0.000	.7906011	.8367091

Note: ATET estimate adjusted for covariates, group effects, and time effects.

[Back to RCS quick](#)

- Now consider a DGP with time-invariant covariates and look at standard errors with and without frequency weights

[Back to RCS quick](#)

Listing 6: rcs_sim_tp2_full.do

```
1 // repeated cross-sectional data, i within s foreach t=1,2,
2 // treatment-assignment at level s, time-invariant covariates
3 // Do not collapse down to s and t data
4 local n1 = 10
5 local n2 = 15
6 local S = 5000
7 local df = 25
8 local dist " (rchi2(`df') - `df')/sqrt(2*`df') "
9 set obs `S'
10 generate s=_n
11 generate alpha_s = rlogistic()
12 generate x1 = `dist' + .2
13 generate x2 = `dist' + .2
14 generate z1 = (alpha_s + `dist')/sqrt(2) + .2
15 generate z2 = (alpha_s + `dist')/sqrt(2) + .2
16 expand 2
17 sort s
18 by s: generate t=_n
19 generate n = cond(t==1, `n1', `n2')
20 expand n
21 sort t s
22 by t s: generate id = (t-1)*100 + _n // id's differ by sample
23 local lam_1 = `dist'
24 local lam_2 = `dist'
25 generate lam_t = cond(t==1, `lam_1', `lam_2')
26 generate d = (.1*x1 + .1*x2 + - 0.15 + alpha_s > 0)
27 generate w = d==1 & t==2
28 generate y1 = w*.8 + 1 + .3*z1 + .5*z2 + alpha_s + lam_t + `dist'
29 generate y0 = 1 + .3*z1 + .5*z2 + alpha_s + lam_t + `dist'
30 generate y = d*y1 + (1-d)*y0
31 save rcs_sim_tp2_full, replace
```

- Draw and describe simulated data
- Note huge number of observations

```
. drop _all
. set seed 12345
. quietly do rcs_sim_tp2_full
. describe
```

Contains data from rcs_sim_tp2_full.dta

Observations: 125,000

Variables: 15

29 Oct 2024 15:13

Variable name	Storage type	Display format	Value label	Variable label
s	float	%9.0g		
alpha_s	float	%9.0g		
x1	float	%9.0g		
x2	float	%9.0g		
z1	float	%9.0g		
z2	float	%9.0g		
t	float	%9.0g		
n	float	%9.0g		
id	float	%9.0g		
lam_t	float	%9.0g		
d	float	%9.0g		
w	float	%9.0g		
y1	float	%9.0g		
y0	float	%9.0g		
y	float	%9.0g		

Sorted by: t s

[Back to RCS quick](#)

Look at distribution of d and sample ATET from potential outcomes

```
. tab d if t==1
```

d	Freq.	Percent	Cum.
0	26,620	53.24	53.24
1	23,380	46.76	100.00
Total	50,000	100.00	

```
. generate te = y1 - y0
```

```
. mean te if d==1, over(t)
```

Mean estimation

Number of obs = 58,450

	Mean	Std. err.	[95% conf. interval]	
c.te@t				
1	-.0080369	.0092062	-.0260811	.0100072
2	.7991729	.0075616	.7843521	.8139938

[Back to RCS quick](#)

Estimate ATET using didregress

```
. didregress (y z1 z2 ) (w) , group(s) time(t)
note: z1 omitted because of collinearity.
note: z2 omitted because of collinearity.
```

Treatment and time information

Time variable: t

Control: w = 0

Treatment: w = 1

		Control	Treatment
Group	s	2662	2338
Time	Minimum	1	2
	Maximum	1	2

Difference-in-differences regression

Number of obs = 125,000

Data type: Repeated cross-sectional

(Std. err. adjusted for 5,000 clusters in s)

y	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
ATET (1 vs 0) ^w	.7872833	.0119152	66.07	0.000	.7639242	.8106423

Note: ATET estimate adjusted for covariates, group effects, and time effects.

. estimates store rcs_twfe

[Back to RCS quick](#)

Collapse the data to observed mean within s for each t

```
. collapse y y0 y1 d w z1 z2 x1 x2 , by(s t)
. generate fw = cond(t==1, 10, 15)
. save rcs_sim_tp2_short, replace
file rcs_sim_tp2_short.dta saved
```

[Back to RCS quick](#)

Point estimate of ATET is identical to full data, SE is much larger

```
. didregress (y z1 z2 ) (w) , group(s) time(t)
```

note: z1 omitted because of collinearity.

note: z2 omitted because of collinearity.

Treatment and time information

Time variable: t

Control: w = 0

Treatment: w = 1

		Control	Treatment
Group	s	2662	2338
Time			
Minimum		1	2
Maximum		1	2

Difference-in-differences regression

Number of obs = 10,000

Data type: Repeated cross-sectional

(Std. err. adjusted for 5,000 clusters in s)

y	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
ATET						
(1 vs 0)	.7872833	.0165126	47.68	0.000	.7549113	.8196552

Note: ATET estimate adjusted for covariates, group effects, and time effects.

[Back to RCS quick](#)

Using frequency weights yields point estimates and SE that are identical to full data

```
. use rcs_sim_tp2_short, clear
. didregress (y z1 z2 ) (w) [fw=fw] , group(s) time(t)
note: z1 omitted because of collinearity.
note: z2 omitted because of collinearity.
```

Treatment and time information

Time variable: t

Control: w = 0

Treatment: w = 1

		Control	Treatment
Group	s	2662	2338
Time	Minimum	1	2
	Maximum	1	2

Difference-in-differences regression

Number of obs = 125,000

Data type: Repeated cross-sectional

(Std. err. adjusted for 5,000 clusters in s)

y		Robust		t	P> t	[95% conf. interval]	
		Coefficient	std. err.				
ATET	w						
(1 vs 0)		.7872833	.0119152	66.07	0.000	.7639242	.8106423

Note: ATET estimate adjusted for covariates, group effects, and time effects.

Simulation to illustrate that fw se is better

Listing 7: rcs_se_sim.do

```
1
2 clear all
3 set seed 1234
4 local R = 2000
5 local qui quietly
6
7 postfile sim r b_full se_full r_full b_nofw se_nofw r_nofw b_fw se_fw r_fw ///
8 using rcs_se_sim, replace
9 forvalues r = 1/`R' {
10     `qui' drop _all
11     `qui' do rcs_sim_tp2_full
12     `qui' didregress (y z1 z2 ) (w) , group(s) time(t) // estimation using full data
13     local b_full = _b[r1vs0.w]
14     local se_full = _se[r1vs0.w]
15     `qui' test _b[r1vs0.w] = .8
16     local r_full = (r(p)<.05)
17
18     `qui' collapse y y0 y1 d w z1 z2 x1 x2 , by(s t)
19     generate fw = cond(t==1, 10, 15)
20
21     `qui' didregress (y z1 z2 ) (w) , group(s) time(t)
22     local b_nofw = _b[r1vs0.w]
23     local se_nofw = _se[r1vs0.w]
24     `qui' test _b[r1vs0.w] = .8
25     local r_nofw = (r(p)<.05)
26
27     `qui' didregress (y z1 z2 ) (w) [fw=fw] , group(s) time(t)
28     local b_fw = _b[r1vs0.w]
29     local se_fw = _se[r1vs0.w]
30     `qui' test _b[r1vs0.w] = .8
```

```

31     local r_fw = (r(p)<.05)
32
33     post sim (`r') (`b_full') (`se_full') (`r_full')    ///
34         (`b_nofw') (`se_nofw') (`r_nofw')              ///
35         (`b_fw') (`se_fw') (`r_fw')
36
37     if `r'/50 == int(`r'/50) {
38         di ".          `r'"
39     }
40     else {
41         di _c " ."
42     }
43
44 }
45
46 postclose sim
47 use rcs_se_sim, clear
48 summarize

```

[Back to RCS quick](#)

Simulation Results

```
. //do rcs_se_sim  
. use rcs_se_sim, clear  
. summarize
```

Variable	Obs	Mean	Std. dev.	Min	Max
r	2,000	1000.5	577.4946	1	2000
b_full	2,000	.8003131	.0114114	.7617437	.8367342
se_full	2,000	.0118024	.0001218	.0113346	.0122309
r_full	2,000	.041	.1983398	0	1
b_nofw	2,000	.8003131	.0114114	.7617437	.8367343
se_nofw	2,000	.0163563	.0001688	.015708	.01695
r_nofw	2,000	.0045	.0669477	0	1
b_fw	2,000	.8003131	.0114114	.7617437	.8367343
se_fw	2,000	.0118024	.0001218	.0113346	.0122309
r_fw	2,000	.041	.1983398	0	1

[Back to RCS quick](#)

NA assumption in a multiperiod common-entry design

- In a multiperiod common-entry-time design with time-invariant covariates, the NA assumption is

$$\mathbf{E}[y_{1,i,t} - y_{0,i,t} | \mathbf{z}_i, d_i = 1] = 0 \text{ for all } t < q$$

where $t = q$ is the common-entry time

- The ATET in the periods before the treatment are 0
The future treatment does not cause a response before it is given
- Treatment assignment is not related to pre-treatment differences in the potential outcomes

[Back to more complicated designs](#)

PT assumption in a multiperiod common-entry-time design

- In a multiperiod common-entry-time design with time-invariant covariates, the PT assumption is

$$\mathbf{E}[y_{0,i,t} - y_{0,i,t-1} | \mathbf{z}_i, d_i] = \mathbf{E}[y_{0,i,t} - y_{0,i,t-1} | \mathbf{z}_i] \quad t \in \{2, \dots, T\}$$

- The trend in the control potential outcome is not related to the treatment process, after conditioning on covariates
- This rules out any relationship between the time-varying observables that affect the control potential outcome and treatment assignment
- Note that it does not rule out a relationship between the post-treatment-time-period time-varying observables that affect the treatment potential outcome and treatment assignment
 - NA rules out any relationship between the pre-treatment-time-period time-varying observables that affect the treatment potential outcome and treatment assignment

Notation for staggered entry with panel data

- Use a version of the notation in [Callaway and Sant'Anna \(2021\)](#) and the notation that we have been using so far
- See individuals i in time period $t \in \{1, \dots, \mathcal{T}\}$
- Let d_i be 1 if person i gets the treatment at some point and 0 if person i is never treated
 - As before, treatment is irreversible
- No one is treated in time 1

[Back to more complicated designs](#)

More notation for staggered entry

- Let G_i be the period in which person i is treated
 - G_i gives the cohort to which i belongs
Think of G_i as the group for i , as well the treatment period
 - For those that are never treated $G_i = 0$
Callaway and Sant'Anna (2021) use $G_i = \infty$
In data and simulations people use $G_i = 0$
 - $g_{min} = \min_{i: G_i > 0} G_i$
 - $g_{max} = \max_i G_i$
 - Let $\tilde{G}$ be the set of time periods in which some units are first treated
 - For those that get the treatment $G_i \in \{g_{min}, \dots, \mathcal{T}\}$
- Define $G_{i,g}$ to be 1 if person i is first treated in period g and 0 is not first treated in period $g \in \{2, \dots, \mathcal{T}\}$
- Every person has a treatment group g

More notation for staggered entry

- $y_{i,t}(0)$ is the not-treated/control potential outcome of i in t
- $y_{i,t}(g)$ is the treated-potential outcome of i in t for those first treated in time g for $g \in \tilde{G}$
- The observed outcome is

$$y_{i,t} = G_{i,0}y_{i,t}(0) + \sum_{g \in \tilde{G}} G_{i,g}y_{i,t}(g)$$

[Back to more complicated designs](#)

- The effects of interest are

$$ATE(t, g) = \mathbf{E}[y_{i,t}(g) - y_{i,t}(0) | G_{i,g} == 1]$$

for each $g \in \tilde{G}$ and for each $t \geq g$

- Because we observe $y_{i,t}(g)$ when $G_{i,g} = 1$ and $t \geq g$

$$\begin{aligned} ATE(t, g) &= \mathbf{E}[y_{i,t}(g) | G_{i,g} = 1] - \mathbf{E}[y_{i,t}(0) | G_{i,g} == 1] \\ &= \mathbf{E}[y_{i,t} | G_{i,g} = 1] - \mathbf{E}[y_{i,t}(0) | G_{i,g} == 1] \end{aligned}$$

- We need assumptions that will allow us to identify and to estimate the counterfactual $\mathbf{E}[y_{i,t}(0) | G_{i,g} == 1]$

[Back to more complicated designs](#)

- [Callaway and Sant'Anna \(2021\)](#) use an overlap assumption, the CS NA assumption and a parallel trends assumption
 - [Callaway and Sant'Anna \(2021\)](#) have two version of the parallel trends assumption
 - One version uses a never-treated group
 - Another version uses a not-yet-treated group
- We only discuss the version that uses a never-treated group, which is similar to the base-case in [Wooldridge \(2021\)](#)

Back to more complicated designs

No anticipation, staggered entry

- CS NA

$$\mathbf{E}[(y_{i,t}(g) - y_{i,t}(0)) | \mathbf{x}_i, G_{i,g} = 1] = 0$$

for $g \in \tilde{G}$ and all $t < g$

- The pre-treatment ATETs are 0

The treatment has no effect prior to implementation

The treatment is not anticipated

- CS NA rules out any relationship between pre-treatment time-varying unobservables that affect any of the potential outcomes and the treatment-assignment process

Back to more complicated designs

CS parallel trends, staggered entry

- CS PT Never treated

$$\mathbf{E}[y_{i,t}(0) - y_{i,t-1}(0) | G_{i,g} = 1] = \mathbf{E}[y_{i,t}(0) - y_{i,t-1}(0) | d_i = 0]$$

for all $g \in \tilde{G}$ and all $t \geq g$

- In treatment periods, the trend for not-treated potential outcome is the same for those get the treatment and those who are never treated
- This rules out any effect of the time-varying unobservables for the untreated potential outcome on the treatment assignment process in the treated periods

[Back to more complicated designs](#)